

LLM-based Conversational AI Therapist for Daily Functioning Screening and Psychotherapeutic Intervention via Everyday Smart Devices

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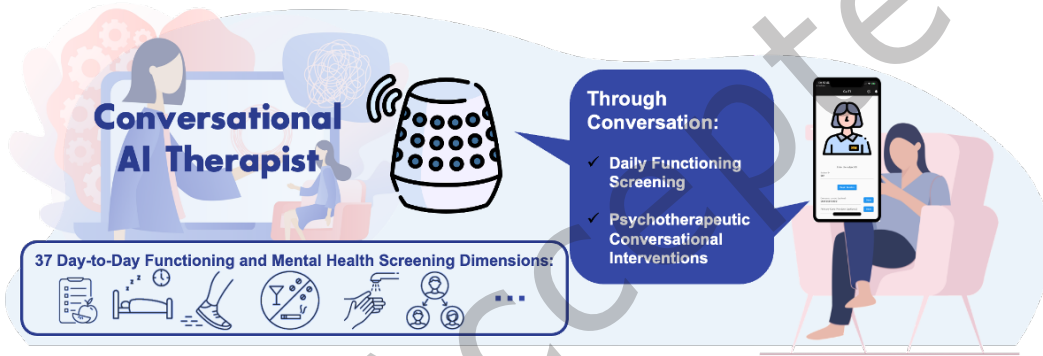


Fig. 1. CaiTI is a conversational “AI therapist” that screens and assesses the day-to-day functioning of the occupant across 37 dimensions and provides appropriate psychotherapeutic interventions and empathic validations depending on the physical and mental status of the occupant.

Despite the global mental health crisis, access to screenings, professionals, and treatments remains high. In collaboration with licensed psychotherapists, we propose a Conversational AI Therapist with psychotherapeutic Interventions (CaiTI), a platform that leverages large language models (LLM)s and smart devices to enable better mental health self-care. CaiTI can screen the day-to-day functioning using natural and psychotherapeutic conversations. CaiTI leverages reinforcement learning to provide personalized conversation flow. CaiTI can accurately understand and interpret user responses. When the user

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needs further attention during the conversation, CaiTI can provide conversational psychotherapeutic interventions, including cognitive behavioral therapy (CBT) and motivational interviewing (MI). Leveraging the datasets prepared by the licensed psychotherapists, we experiment and microbenchmark various LLMs' performance in tasks along CaiTI's conversation flow and discuss their strengths and weaknesses. With the psychotherapists, we implement CaiTI and conduct 14-day and 24-week studies. The study results, validated by therapists, demonstrate that CaiTI can converse with users naturally, accurately understand and interpret user responses, and provide psychotherapeutic interventions appropriately and effectively. We showcase the potential of CaiTI LLMs to assist the mental therapy diagnosis and treatment and improve day-to-day functioning screening and precautionary psychotherapeutic intervention systems.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Computing methodologies** → *Discourse, dialogue and pragmatics*.

Additional Key Words and Phrases: Large Language Models (LLMs), Foundation Models, AI therapist, Psychotherapy, Everyday Smart Devices, Cognitive Behavioral Therapy, Motivational Interviewing

1 INTRODUCTION

Maintaining physical and mental health is crucial for quality of life, particularly for those living alone, experiencing early signs of mental illness, or requiring daily assistance. COVID-19 significantly impacted global mental health with high barriers to accessing mental health screenings and treatments, including home care [42, 87]. There are a variety of smart wearables and smart devices to monitor physical and mental health [59, 62, 104]. Human-Computer Interaction (HCI) researchers are actively working to improve wellness care for the general public and vulnerable population groups [57, 69, 89]. Furthermore, with the growth of the Internet of Things (IoT) devices, the smart speaker market volume reached 200 million and there are more than 6.84 billion smartphones worldwide in 2023 [35, 86]. Recent advances in artificial intelligence (AI) and large language models (LLMs) further expanded the possibilities for intelligent health-oriented applications [29, 62, 64, 93, 102].

While existing research primarily focuses on understanding emotional states or affective states as indicators of mental well-being [103], therapists generally require more knowledge about patients' daily activities and behaviors to accurately assess mental health [16, 41]. Therapists often rely on assessments such as the Daily Living Activities-20 (DLA-20) and the Global Assessment of Functioning (GAF) to screen day-to-day functions and mental health status [27, 37, 39, 59]. Most existing research efforts focus on screening for physical and mental well-being, with few addressing psychotherapeutic interventions. Psychotherapy refers to a range of interventions based on psychological theories and principles to address emotional and behavioral issues that impact mental health [28]. [63] and [103] propose conversational systems that provide preliminary consolation. While conversational systems and evidence-based treatments like Motivational Interviewing (MI) [61], Cognitive Behavioral Therapy (CBT) [15], and Dialectical Behavior Therapy (DBT) [73] have been proposed, many lack personalization or user understanding [75, 77].

Although AI chatbots like ChatGPT show promise in addressing mental health concerns [3, 21], they often suffer from performance decline over time and limitations in psychotherapeutic considerations [4]. LLMs also fall short in autonomous planning and reasoning [44]. Additionally, mental health applications see low usage rates in clinical settings [22, 90]. Smartphone-based tools may not be user-friendly for individuals with memory or vision impairments, especially the elderly [58]. As such, there is growing interest in exploring objective activity detection through ambient sensing and voice-based chatbots as more inclusive and effective approaches to mental health support.

Considering these limitations and opportunities, in collaboration with 4 licensed psychotherapists from a major mental health counseling institution with thousands of clients, we propose *CaiTI, a conversational AI therapist that takes advantage of widely-owned smart devices for continuous screening of physical and mental health in a privacy-aware manner, while employing psychotherapeutic interventions* (Figure 1). Our collaborating psychotherapists have identified several design requirements for CaiTI that can facilitate mental health self-care

and assist in psychotherapeutic treatment for individuals: (i) provide comprehensive day-to-day functioning screenings and employ evidence-based psychotherapeutic interventions; (ii) facilitate natural conversation flow; (iii) ensure the quality of care by enabling the system to intelligently interpret user responses and, if necessary, guide the dialogue back toward the psychotherapeutic objectives when the user's responses deviate; and (iv) the conversation format (using smartphones/smart speakers) should take into consideration individuals with visual impairments.

Realizing such a system poses several challenges. Primarily, the system must *fit* within the users' lifestyles and habits, utilizing devices that users already own and prefer. It should facilitate communication through the user's preferred modes—be it verbal or textual—while ensuring *comprehensive screening* and delivering *effective psychotherapeutic interventions* in a *privacy-aware* manner. Additionally, it is imperative that the system is *easy to use* for all individuals, regardless of their technical proficiency. Furthermore, the LLMs in CaiTI should effectively deliver conversational psychotherapy and must be carefully designed to both be *user-friendly/accessible* and *capable* of understanding, reasoning, and responding to an infinitely diverse number of user responses (including both YES/NO answers and open-ended responses). The design of CaiTI must effectively manage varied responses, translate therapists' empirical techniques into a quality-controlled logical flow, and incorporate a recommendation system that dynamically personalizes itself to each user.

Building upon near-ubiquitous smart devices, CaiTI combines AI techniques, including LLMs, reinforcement learning (RL), and human-computer interaction (HCI) approaches with professional experiences from licensed psychotherapists. CaiTI *screens the user along the 37 dimensions of day-to-day functioning* proposed in [63] by conversing naturally with users with open-ended questions. CaiTI understands verbal and textual responses and activities of the user and employs *conversation-based psychotherapeutic interventions*. To summarize, the main contributions of this paper include:

- CaiTI, an LLM-based conversational “AI therapist” that screens and analyzes the day-to-day functioning of users across 37 dimensions. Using the screening results, CaiTI provides appropriate empathic validations and psychotherapies depending on the physical and mental status of the user. CaiTI is accessible through widely available smart devices, including smartphones, computers, and smart speakers, and offers a versatile solution catering to the diverse requirements of the users, whether they are indoors or outdoors.
- To realize more intelligent and *friendly* human-device interaction, we leverage RL to personalize each user's conversation experience during screening in an adaptive manner. CaiTI prioritizes the dimensions that concern psychotherapists more about each user based on their historical responses and brings up the dimensions in the order of priority during the conversation.
- We design the conversation architecture of CaiTI with the therapists, which effectively incorporates Motivational Interviewing (MI) and Cognitive Behavioral Therapy (CBT) – two commonly used psychotherapeutic interventions administered by psychotherapists – to provide *Psychotherapeutic Conversational Intervention* in a natural way that closely mirrors the therapists' actual practices.
- To ensure the quality of care and effectiveness of the psychotherapy process and avoid the propagation of biases in AI algorithms and LLMs, CaiTI incorporates multiple task-specific LLM-based **Reasoners**, **Guides**, and **Validator** during the psychotherapy process. Leveraging the task-specific conversation datasets prepared and labeled by the licensed psychotherapists, we experiment and microbenchmark the performance of different GPT- and Llama 2-based LLMs with few-shot prompts or fine-tuning in performing tasks along CaiTI's. The datasets prepared by the therapists and the few-shot prompts we designed with the therapists are available at: https://github.com/Columbia-ICSL/CaiTI_dataset.
- In collaboration with licensed psychotherapists, we design, implement, and deploy a proof-of-concept prototype of CaiTI. Through real-world deployments with 20 subjects for up to 24 weeks, we demonstrate that CaiTI can

accurately assess the user's physical and mental status and provide appropriate and effective psychotherapeutic interventions. CaiTI has received positive feedback, endorsements, and validation from both licensed psychotherapists and subjects.

To the best of our knowledge, CaiTI is the first conversational "AI therapist" system that leverages smart home devices and LLMs to mimic the psychotherapists' actual practices in clinical sessions and provides continuous monitoring and interaction with the integration of psychotherapies (MI and CBT).

2 PSYCHOLOGICAL BACKGROUND

2.1 Psychological Assessment

People who experience mental health adjustment issues and disorders tend to face diminished capacity in professional or academic performance, maintaining social relationships, and self-care [16, 41]. Traditional screening tools, such as the Mental Status Examination (MSE), require clinicians to observe and assess people's daily functioning, such as physical appearance and presentation, social interaction behaviors, and emotional expression [92]. Other widely used diagnostic assessments, such as the Adult ADHD Self-Report Scale (ASRS-v1.1), the Patient Health Questionnaire-9 (PHQ-9) for depression, and the General Anxiety Disorder-7 (GAD-7), which provide more specific screening options for specific mental health diagnoses, often include questions or items assessing daily functioning [33, 47]. For example, the PHQ-9 includes assessments of mood, sleep hygiene, and eating habits [47].

There are several psychological measurements designed to examine the day-to-day functioning of individuals to evaluate their mental health well-being, such as DLA-20 and GAF [32, 63, 78]. DLA, which was developed to evaluate aspects of daily functioning affected by mental illnesses, includes 20 major categories for daily functioning. These categories include interpersonal communication, family relationships, personal hygiene, time management, and productivity at work [78]. On the other hand, GAF, which was introduced in DSM-IV, employs an ordinal scale to evaluate an individual's overall level of functioning [32]. A lower GAF score indicates the presence of more significant symptoms and difficulties in social, occupational, and psychological functioning.

2.2 Psychotherapeutic Interventions

Clinicians using evidence-based practices (EBP) in psychology to guide interventions and treatment plans, taking into account relevant research on their clinical practices, have found that the use of EBP helps improve the quality and accountability of clinical practices [65, 85]. Some commonly used EBP include CBT, acceptance and commitment therapy (ACT), DBT, and MI. CBT is one of the most popular and commonly used psychological interventions. It focuses on challenging one's cognitive distortions and subsequent behaviors to reduce existing mental health symptoms and improve overall mental well-being [9, 15]. CBT is found to be effective in a variety of diagnoses, such as mood disorders, Attention-deficit/hyperactivity disorder (ADHD), eating disorders, Obsessive-compulsive disorder, and Post-traumatic stress disorder [26, 31, 34, 36, 40, 74, 96]. CBT also shows promising results in preventative care that may not be tied to a specific diagnosis. It has been effective in various settings, including medical, work, and school environments [55, 60, 88]. However, despite the abundant evidence of its effectiveness, CBT is associated with a high nonresponse rate, attributed to participants' low motivation [13]. During CBT, therapists assess the validity and utility of participants' responses to understand their thought patterns and beliefs accurately [84]. Such an assessment involves identifying, challenging, and reframing cognitive distortions, such as overgeneralization, emotional reasoning, all-or-nothing thinking, and catastrophizing [20]. CBT usually consists of the following steps:

- (1) **Identify the Situation/Issue:** Start by clearly identifying the situation or issue you want to work on.

- (2) **Recognize Negative Thoughts:** Think about the thoughts that go through your mind when you experience this issue. These are often automatic or subconscious thoughts that may be irrational or unhelpful. They can be self-critical, overly pessimistic, or unrealistic.
- (3) **Challenge Negative Thoughts:** Challenge means questioning the validity of these thoughts. Are there alternative, more balanced, or rational thoughts that might be more helpful in the situation?
- (4) **Reframe Thoughts and Situations:** Try to reframe your unhelpful thoughts and situations into more balanced, realistic, and constructive ones. This process is about changing the way you think about the situation, which can lead to changes in your emotions and behaviors.

To address issues related to low motivation, researchers have suggested using MI as a complementary approach alongside CBT [14, 52, 61]. There are four techniques to effectively implement MI [56]:

- (1) **Open-ended questions:** Encouraging elaboration on responses, asking for examples, or exploring the implications of what's been shared;
- (2) **Affirmations:** State strengths and help feel that changes are possible;
- (3) **Reflective listening:** (i) Simple reflection: repeating what the client has said, using slightly different words or phrases; (ii) Reframe reflection: listening to the client's statements and then reflecting them back in a way that presents a new perspective or interpretation; and (iii) Affective reflection: recognizing, understanding, and reflecting back the emotional content of what the client expresses;
- (4) **Summaries:** Use summaries not only to encapsulate discussions but also to highlight progress.

MI is an evidence-based practice for substance use disorders and other addiction issues [6, 12]. It is also found to be effective in helping people adapt to various situations, such as managing diabetes [23, 25, 45]. Growing research has shown that the combination of CBT and MI shows effectiveness in a variety of populations and for mental health adjustments [14, 45, 52, 54].

3 RELATED WORK

3.1 Mental Wellness Self-Screening and Self-Care

There are various methods for mental health self-screening [18, 48]. While online help-seeking is preferred by many individuals [38], these tools provide a limited assessment based on closed-ended questions, potentially leading to omitting important details typically obtained from open-ended questions or in-person interactions [1, 79]. Besides the online tools, [51] proposed an AI-based self-administered online web-browser-based mental status examination (MSE). Recently, the Experience Sampling Method (ESM) has been widely adopted by HCI researchers for various physical and mental health screening. ESM can be done automatically by sensors or by repeatedly prompting users to answer questions in their normal environments. For example, ESM is used for self-reporting Parkinson's Disease symptoms, chronic pain, designing health technologies for Bipolar Disorder, etc [7, 53, 94]. Most wellness self-screening methods in the literature employ close-ended questions and expect similarly structured responses from the user, whereas CaiTI utilizes open-ended questions, allowing users to freely discuss any topic. Additionally, these methods typically focus only on specific dimensions of day-to-day functioning or mental disorders, rather than conducting a comprehensive screening.

3.2 LLM-based Healthcare and Mental Healthcare

Large Language Models (LLMs) are pre-trained on vast datasets, which equip them with significant prior knowledge and enhanced reasoning skills. The recent state-of-the-art models, including GPT-4 [67], GPT-3 [66], Claude-3 [11], and Gemini 1.5 [70], exhibit strong capability in reasoning over text. Consequently, recent research increasingly employs LLMs alongside various language, vision, or multimodal models to enable advanced applications in various domains without the need for additional training [30, 80, 100]. For example, IdealGPT combines two LLMs (GPT) with a vision-and-language model to enable a framework that iteratively decomposes

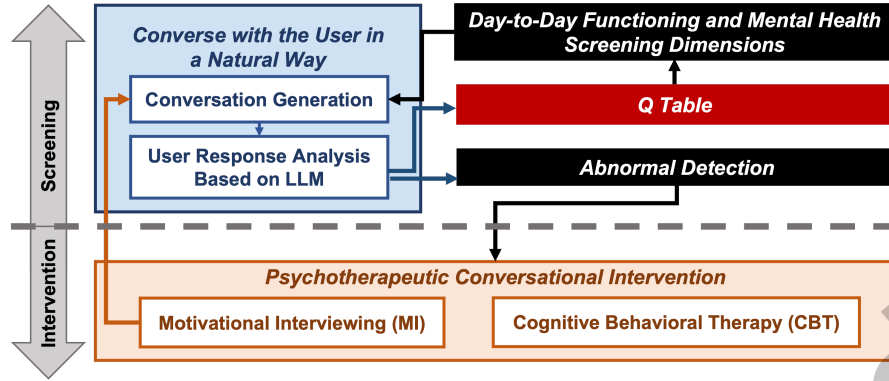


Fig. 2. The system architecture of CaiTI consists of two main functionalities: 1) day-to-day functioning screening through natural conversation and 2) precautionary psychotherapeutic conversational interventions.

vision-and-language reasoning, where the two LLMs are treated as Questioner and Reasoner [101]. Additionally, research has shown that LLMs possess the ability to reason with and interpret IoT sensor data [97].

Recently, transformer-based large language foundation models, such as GPT-4 [19], PaLM 2 [10], and LLaMA2 [91], have demonstrated superior performance across various medical-related NLP tasks. LLMs are used to enable various general healthcare applications. [95] showed that GPT-4 has the potential to help drive medical innovation, from aiding with patient discharge notes, summarizing recent clinical trials, and providing information on ethical guidelines. Moreover, Google introduced Med-PaLM and Med-PaLM 2 [82, 83], LLMs explicitly tailored for the medical domain, providing high-quality responses to medical inquiries.

Various works also exploit and evaluate the performance of LLMs for mental status classification and assessment. Researchers leveraged LLMs for mental health prediction via online text data and evaluated the capabilities of multiple LLMs on various mental health prediction tasks via online text data [72, 98]. MentalLLaMA was proposed to analyze and interpret mental health content from social media posts using LLMs [99]. In addition, [43] leverages RoBERTa [50] and Llama-65b [91] in the system for classifying psychiatric disorder, major depressive disorder, self-rated depression, and self-rated anxiety based on time-series multimodal features.

In addition to assessing and classifying the mental status of the user, researchers have investigated providing psychological consultations. For example, [63] leveraged GPT-3 to construct a home-based AI therapist that detects abnormalities in mental status and daily functioning and generates responses to console users. [49] proposed an AI-based assistive tool leveraging the WenZhong model, a pre-trained model trained on a Chinese corpus for question-answering in psychological consultation settings [49]. Researchers also investigate the potential of ChatGPT in powering chatbots to simulate the conversations between psychiatrists and mentally disordered patients [24].

Only a few works in the literature focus on using LLM for MI or CBT, developing these psychotherapy systems, and evaluating them in real-world scenarios. [46] developed a GPT3.5-powered prompt-engineered socially assistive robot (SAR) that guides participants through interactive CBT at-home exercises. Their findings suggest that SAR-guided LLM-powered CBT may yield comparable effectiveness to traditional worksheet methods. However, this study solely focused on employing an LLM-based approach to simulate traditional worksheet-based CBT, without thoroughly examining the validity of user responses to the CBT exercises or ensuring users effectively engaged with the CBT.

4 SYSTEM ARCHITECTURE

Considering the design requirements presented in Section 1, CaiTI includes two main functionalities: *day-to-day functioning screening* and *precautionary psychotherapeutic conversational interventions* as shown in Figure 2. We adopt the 37 dimensions for day-to-day functioning screening proposed in [63]. For screening, *Converse with the User in a Natural Way* consists of open-ended question generations and semantic analysis of user responses based on LLM. To facilitate *precautionary psychotherapeutic conversational interventions*, following psychotherapists' guidance, CaiTI effectively integrates the motivational interviewing (MI) and cognitive behavior therapy (CBT) processes into *Psychotherapeutic Conversational Intervention*. Considering the characteristics of MI and CBT, and the actual ways in which the therapists perform during clinical sessions, various MI techniques introduced in Section 2.2 are applied in different scenarios during the conversation, while the four-step CBT is performed at the end of each conversation session.

Each activity screened through conversation sensing results in a **(Dimension, Score)** pair. Therapists set 3 classes for **Score** based on their clinical practices ($\text{Score} \in \{0, 1, 2\}$), where (i) a score of 0 indicates that the user performs well in this dimension, (ii) a score of 1 indicates that the user has some problems in this dimension, but no immediate action is needed, and (iii) a score of 2 indicates a need for heightened attention from healthcare providers. Figure 3 shows the flow diagram of CaiTI's components, and CaiTI stores the historical user data on the front-end devices owned by the user. Due to privacy concerns, CaiTI only conducts semantic analysis on the text of user input, although speech audio is informative [76].

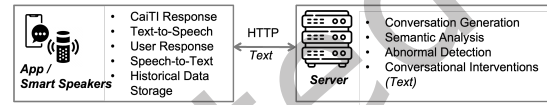


Fig. 3. The workflow of CaiTI, including the connections between the main components and the locations where different functionalities take place.

4.1 Conversation Principles and Underlying Rationale

Converse with the User in a Natural Way and the *Psychotherapeutic Conversational Intervention* modules of CaiTI are closely related to each other when CaiTI converses with the user. Based on the psychotherapists' experience in dealing with thousands of clients, several factors are considered to shape the conversation process of CaiTI. First of all, when the therapist asks a question, some clients express a lot, while others do not respond to the question but talk about other things (related to other dimensions). In addition, not all clients are patient enough to go through all the dimensions that the therapist wants to check. Psychotherapists usually start to check on the dimensions that the clients didn't do well in previous sessions and are more important for assessment. If clients have a problem in a dimension, the therapists usually follow up to hear more about this dimension and provide quick counseling and therapy addressing the specific issue. This mirrors the psychotherapist's tendency to focus on one problematic dimension extensively rather than treating multiple dimensions at once.

Taking the professional experiences and common practices of the psychotherapists into consideration, to converse with the user in an efficient, intelligent, and natural way to screen physical and mental health status, CaiTI's conversation process follows four guidelines:

- (1) *Prioritize questions intelligently*: CaiTI starts with the dimensions that concern therapists more, while personalizing the priority to each user and formulating questions based on their historical responses.
- (2) *Understand the user input better*: CaiTI checks if the user answers the question asked (*Dimension_N*), understands how well the user performs in this dimension and decides if follow-up questions and conversational interventions are needed.
- (3) *Obtain more information through minimal questioning*: CaiTI maps each user response to all possible dimensions to avoid redundant questions.

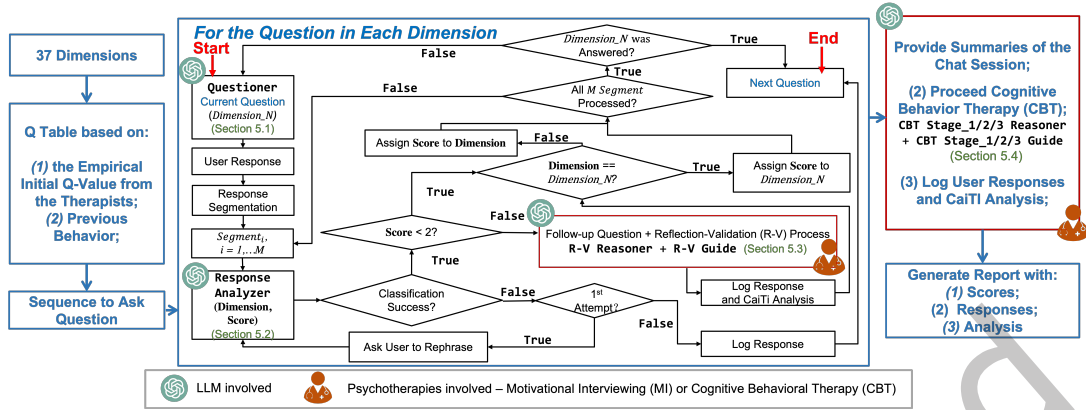


Fig. 4. The process to converse with the user naturally to screen day-to-day functioning and provide psychotherapies through the conversation. In particular, *MI* therapies are conducted throughout the conversation, while *CBT* proceeds at the end of the conversation session.

- (4) *Guarantee the quality of psychotherapies*: CaiTI intelligently interprets and reasons the user responses and, when needed, it steers and guides the conversation back to the psychotherapeutic goals if the user's answers stray.

4.2 Conversation Generation, Analysis, and Psychotherapeutic Intervention

Figure 4 shows CaiTI's process to converse with the user. Generally, *MI* therapies are conducted throughout the conversation, while *CBT* proceeds at the end of the conversation session. There are four modules in which the LLMs are involved: **CaiTI Questioner**, **Response Analyzer**, **reflection-validation (R-V)** process, and **CBT** process.

In particular, CaiTI asks one question for each dimension if CaiTI does not obtain any information in the dimension from the user's previous responses. A model-free reinforcement learning algorithm, Q-learning, is used to decide the action (i.e., the next question) in the current state (i.e., the current question). For each dimension (*Dimension_N*), **CaiTI Questioner** formulates the question and uses the text-to-speech method to converse with the user through the front-end device. The front-end device generates the text of the user response (speech-to-text conversion is used if the user has voice input). The detailed implementation of the front-end device is described in Section 6.3.

CaiTI expects the user to chat freely with it and can deal with open-ended responses. CaiTI performs segmentation on the user response into *M* Segment(s). For each Segment, a LLM-based **Response Analyzer**, described in Section 5.2, is used to classify the Segment into (**Dimension**, **Score**). If CaiTI fails to classify the Segment into the format of (**Dimension**, **Score**), it asks the user to rephrase the answer. CaiTI logs the user response if CaiTI still fails to classify the rephrased Segment into the format of (**Dimension**, **Score**). Otherwise, CaiTI checks the **Score** no matter if this Segment is answering the question asked by CaiTI or not. If the user needs more attention in this dimension (**Score** = 2), CaiTI proceeds with a reflection-validation (R-V) process starting with asking for more information starting with a *simple reflection* in MI. An example of this process is presented in Figure 5.

The R-V process is described in detail in Section 5.3 and demonstrated in Figure 8. To ensure the user provides follow-up information in the right direction, an **R-V Reasoner** and an **R-V Guide** are deployed. Based on the user response to the original and follow-up questions, CaiTI provides validation, which includes *affective reflection* and *affirmations* in MI. Then, CaiTI assigns the **Score** to **Dimension**. After handling all Segment(s), CaiTI verifies whether the user does respond to the question asked by CaiTI in *Dimension_N*. In cases where a user does not

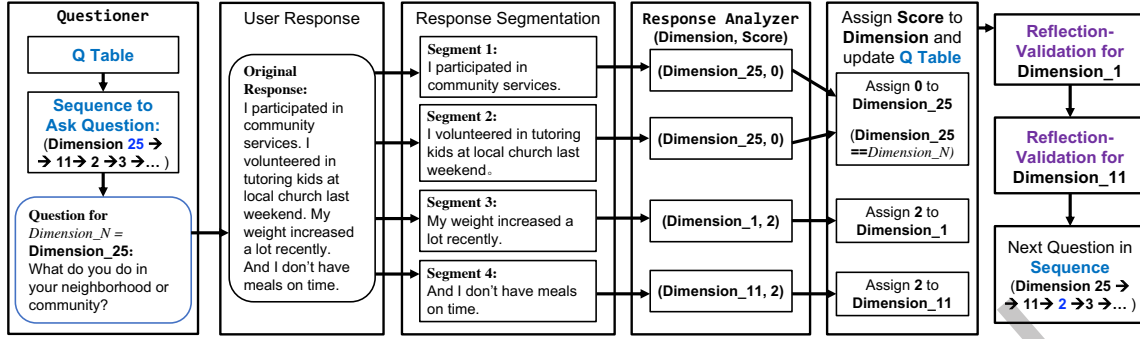


Fig. 5. The example process of how CaiTI follows the sequence based on the Q-table, segments the user response, classifies the *Segment* into the format of **(Dimension, Score)**, and proceeds with reflection-validation (R-V) process.

answer the question asked by CaiTI (*Dimension_N*) but talks about something else, CaiTI asks the question in *Dimension_N* again.

After CaiTI enumerates all dimensions or the user wants to stop the session, CaiTI provides a summary of the chat session and asks the user to choose a dimension to work on for the CBT process. This CBT process includes the four steps outlined in Section 2.2. In particular, CaiTI identifies the situation and issue in the dimension the user chose based on the conversation history. Then, CaiTI leads the user to recognize (**CBT Stage_1**), challenge (**CBT Stage_2**), and reframe (**CBT Stage_3**) the negative thoughts in this situation. To ensure the effectiveness and quality of the CBT process, each CBT stage contains a **Reasoner** and a **Guide** (see Section 5.4).

At the end, CaiTI generates a report that follows the same format as the therapists' notes during their treatment sessions. Appendix A reports the details for the 37 dimensions used for day-to-day functioning screening, example questions from CaiTI, and sample responses from the users. Figure 6 shows the smartphone interface for CaiTI's conversational chatbot, where the various psychotherapeutic interventions applied during different stages of the conversation are annotated.

As CaiTI provides comprehensive daily functioning screening, as presented in Appendix A, some of the dimensions, such as law-abiding, might be sensitive or uncomfortable for users. Therefore, CaiTI offers the option for users to manually select the dimensions to work on with the smartphone interface shown in Figure 7.

5 METHOD AND LLM MICROBENCHMARKS

The methods and LLMs leveraged in: (i) CaiTI's **Questioner**, (ii) **Response Analyzer**, and (iii) task-specific **Reasoners**, **Guides**, and **Validator** during the psychotherapies (MI and CBT processes) are introduced in this section. To prevent the propagation of flaws or biases in LLMs, which may lead to ineffective or potentially harmful psychotherapy intervention, instead of leveraging models to handle all tasks during the psychotherapy process, CaiTI divides the tasks and employs different LLM models to specifically handle each subtask.

Additionally, we present and discuss microbenchmarks comparing different GPT-based methods GPT-4(gpt-4) [67] and GPT-3.5 Turbo (gpt-3.5-turbo-16k) [66] with Llama 2-based methods [91]. In particular, GPT-4, GPT-3.5 Turbo, Llama-2 13b, and Llama-2 7b possess parameter counts of over a trillion, more than one hundred billion, thirteen billion, and seven billion [98]. To carry out the microbenchmarks for LLMs, *datasets were prepared and labeled by four licensed psychotherapists*. These datasets were constructed based on their clinical experiences with clients, ensuring relevance and applicability to real-world therapeutic scenarios.

Considering the training dataset size for each task provided by the therapists, under the guidance from the therapists, we predominantly use *few-shot* prompting the system content in the *chat completion* in these LLMs to achieve the desired functions. Each prompt outlines: (i) the objectives; (ii) the information to be included

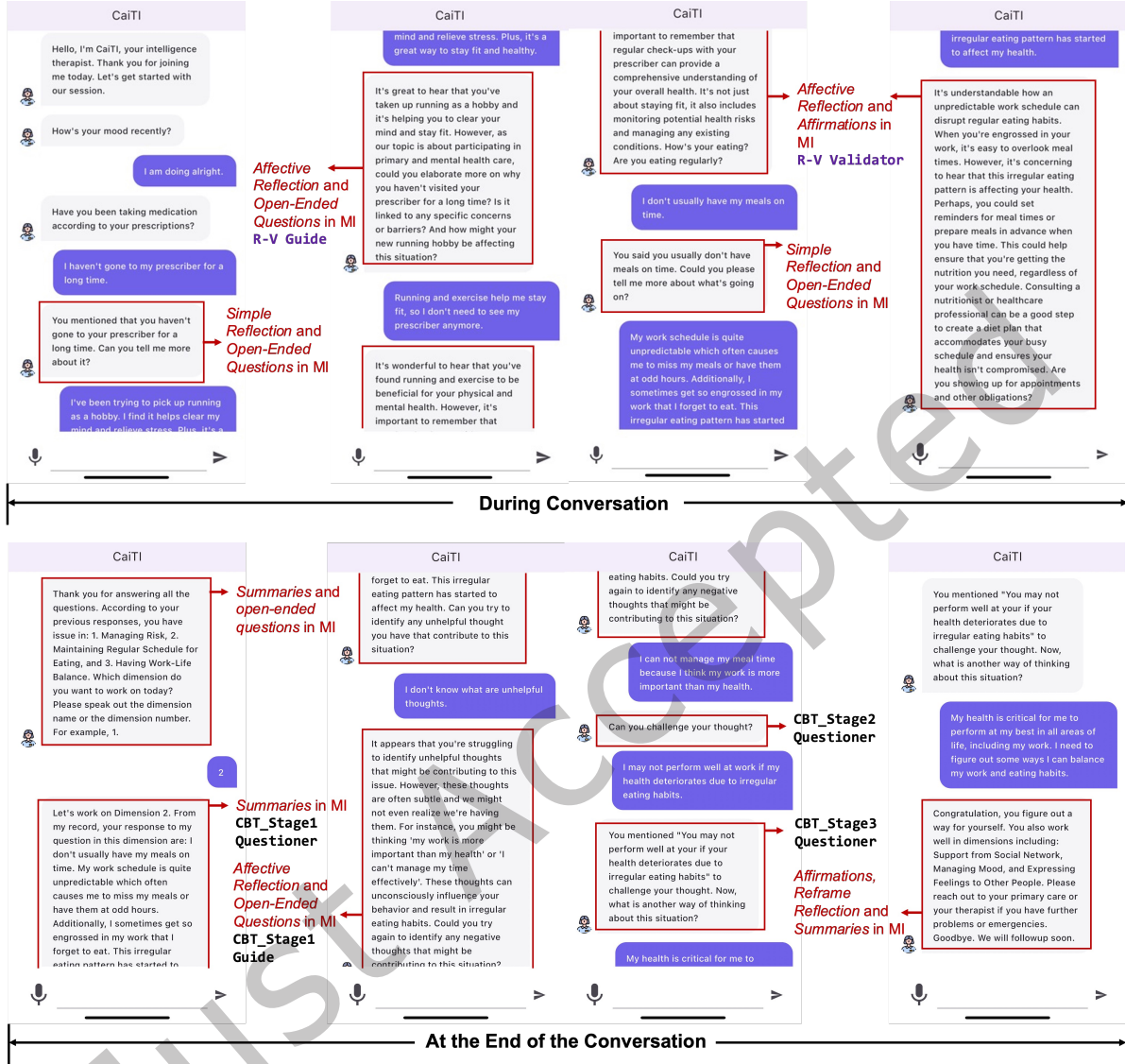


Fig. 6. An example conversation using CaiTi's customized smartphone platform. The MI and CBT embedded in the conversation are annotated.

in user content; and (iii) the desired goal and response format. The response format for **Reasoners** will be "Decision: 0/1", while it is "Analysis: XXX" for **Guides** and **Validator**. For **Reasoners**, **Guides**, and **Validator**, the prompt includes 3-4 examples encompassing user content alongside corresponding system responses that adhere to the specified format. The examples for **Response Analyzer** are slightly different and illustrated in Section 5.2. We set the temperature as 0.7 in the LLMs to achieve varied rephrasings of the questions while maintaining certain constraints. The same prompt and hyperparameters are used for different LLMs. We fine-tune a GPT-3.5 Turbo model for **Response Analyzer**. The therapist also labeled and analyzed the output of **Guides**

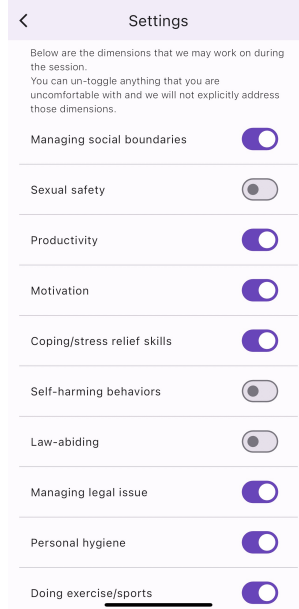


Fig. 7. The user interface where the user can manually select the dimensions to work on.

and **Validator**. We have open-sourced the prompts we constructed, as well as the datasets created by the psychotherapists. We did not conduct microbenchmark tests on basic LLM tasks such as the Rephraser and ReflectiveSummarizer. The former task involves structural rather than semantic rephrasing, while the latter repeats and converts statements from the first person to the third person.

5.1 CaiTI's Questioner

CaiTI's **Questioner** drives the conversations based on Epsilon-Greedy Q-learning and a GPT-based "Rephraser". To make the conversation more natural, psychotherapists provide a set of questions they typically ask in each dimension (7 to 11 sample questions). We prompt a GPT-4-based Rephraser [67] to rephrase these questions (structurally instead of semantically) when asking questions.

Each dimension has one related question. The Q-learning agent has 39 states (37 questions, start, and end). We set the learning rate and discount factor to 0.1 and 0.9, respectively. The probability of selecting the best action is set as $\epsilon = 0.9$. The therapists determine the initial Q-values for the Q-table based on their empirical evaluation of the "importance" of the dimensions. The Q-value represents the expected future rewards that can be obtained by taking a given action (next question) in a given state (current question). The **Score**, based on the analysis of the user's response, represents the reward earned in that state.

5.2 Response Analyzer

When the user responds to the question asked by CaiTI, CaiTI first segments the response into individual sentences (*Segments*). For each segmented sentence, CaiTI classifies it into (**Dimension, Score**), where there are 37 dimensions and 3 scores (**Score** $\in \{0, 1, 2\}$) – a total number of 111 classes. In addition, we define 5-class general responses to express *Yes*, *No*, *Maybe*, *Question*, and *Stop* (e.g., "Yes", "I don't know", "Stop", "Maybe", and "I don't understand your question") as well as a mapping table between the **Scores** and general responses for

Table 1. **Score** and **Dimension** assignment performance comparisons for **Response Analyzer** using different LLM models.

	Score Accuracy	Dimension Accuracy	General Response Accuracy
Fine-tune GPT-3.5-Turbo	95.58%	96%	93.75%
Prompt GPT-4	94.43%	95.14%	95.85%
Prompt GPT-3.5 Turbo	92.81%	94.14%	96.14%
Prompt Llama-2-13b	54.39%	59.42%	63.33%
Prompt Llama-2-7b	55.11%	48.63%	53.33%

each dimension. For example, the **Score** of “Yes” is 0 to the question “Are you showing up for work or school?” in *Managing Work/School*, while it is 2 to the question “Do you often drink alone?” in *Alcohol Abuse*.

Since CaiTI asks open-ended questions, user responses are infinitely diverse (YES/NO answers or open-ended responses). With a **Score** of 2, CaiTI will conduct the psychotherapeutic conversational intervention. Otherwise, CaiTI will ask the next question based on the Q-table. When CaiTI meets out-of-context responses, it asks the user to rephrase and follow the process illustrated in Figure 4.

5.2.1 Microbenchmark – Response Analyzer. To the best of our knowledge, no dataset exists with responses to these questions in the 37 dimensions. Therefore, psychotherapists create a dataset, which includes: (i) 6,950 user responses sample with the (**Dimension, Score**) labeled by the therapists, and (ii) 300 5-class general responses to express *Yes, No, Maybe, Question, and Stop*. Note that one user response may have one or more (**Dimension, Score**). As such, there are 7,000 (**Dimension, Score**) for the 6,950 responses. The number of responses per dimension is 103 to 177.

The datasets are split into 90% for training and 10% for the testing sets to fine-tune and evaluate the GPT-3.5 Turbo model (gpt-3.5-turbo is recommended by OpenAI for fine-tuning tasks [68]). For a fair comparison, the same testing set is used to evaluate the performance of the LLMs with *few-shot* prompt. Therapists select 2-5 examples for each score in each dimension from the training set to construct the *few-shot* system content prompt. The constructed prompt contains around 8,140 tokens. As shown in Table 1, the prompt GPT-4 model demonstrates performance comparable to the fine-tuned GPT-3.5-Turbo model. Llama-based models perform less ideally in correctly classifying the **Dimension** and **Score**. In particular, Llama-based models have issues identifying the responses that don’t need further attention (**Score** = 0 and 1).

5.3 Reflection-Validation Reasoner, Guide, and Validator

As depicted in Section 4.2 and Figure 4, during the conversation screening process, if the user’s response includes situations that require further attention (with **Score** = 2), CaiTI will pose a follow-up question to gather more information about the contributing factors to this situation. The follow-up question would start with the *simple reflection* in MI, a technique where the psychotherapist or counselor mirrors what the client has said. A GPT-4-based ReflectiveSummarizer is prompted to provide the *simple reflection*, which essentially rephrases or repeats the client’s own words, altering any self-references from the first person to the third person [67].

The follow-up question initiates a reflection-validation (R-V) process to provide effective empathic validation and support. Given that follow-up responses from the user are open-ended and have infinite possibilities, CaiTI incorporates a **R-V Reasoner** to determine whether the follow-up response is related to the original response or the question asked in the current state. As illustrated in Scenario 1 in Figure 8, CaiTI will offer empathic validation if the user provides a valid follow-up response. Otherwise, the **R-V Guide** will assist the user in providing a follow-up response that more accurately describes the situation at hand before proceeding to empathic validation (Scenario 2 in Figure 8).

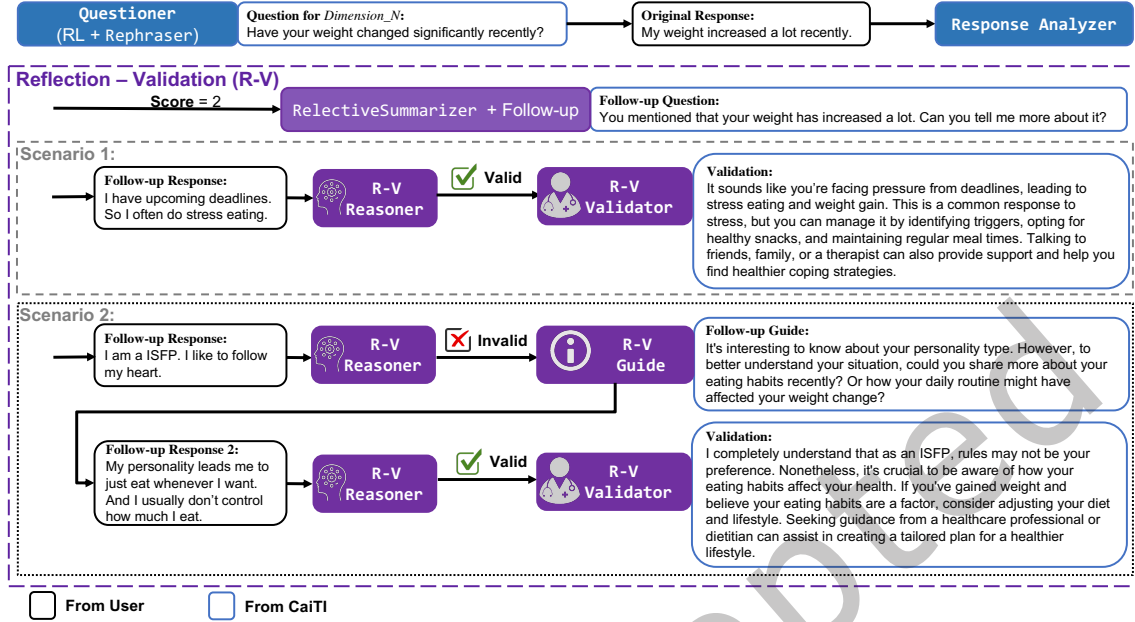


Fig. 8. The reflection-validation (R-V) pipeline illustrates how the *R-V Reasoner* is utilized for reasoning when the user provides a valid follow-up response that necessitates further attention, including two example scenarios: Scenario 1, where the user provides a valid follow-up response on the first attempt; and Scenario 2, where the user initially fails to provide a valid follow-up response, and CaiTI guides the user towards providing valid follow-up responses.

With a valid follow-up response, a **R-V Validator** is used to provide empathic validation and support to the user, which incorporates the *affective reflection* and *affirmation* techniques in MI. These two techniques aim at demonstrating empathy and understanding of the client's feelings, creating a supportive environment that validates the client's experiences and feelings, and fostering a deeper connection.

5.3.1 Microbenchmark – Reflection-Validation Reasoner, Guide, and Validator. Therapists provide us with 593 example follow-up responses for the original responses that need further attention (**Score = 2**). Within these 593 responses, 244 follow-up responses are invalid (1: Invalid), which trigger **R-V Guides** before prompting the **R-V Validators**. In addition, 6 examples for **R-V Guides** and **R-V Validators** are given by the psychotherapists to construct the prompts. As the output of **R-V Guide** and **R-V Validator** are open-ended, therapists label the 244 **R-V Guides** and 593 **R-V Validators** generated by each of the 4 prompt-based LLM models.

Table 2 presents the performance metrics of **R-V Reasoner**, **R-V Guide**, and **R-V Validator** across the four LLM models, where we prompted the system content in the *chat completion* functionality. The data presented in Table 2 illustrate that methods based on Llama demonstrate inferior reasoning capabilities in assessing the validity of a follow-up response in relation to the original question and response within the given context. Typically, Llama-based approaches are prone to inaccurately classifying valid follow-up responses as invalid. Additionally, the GPT-based method demonstrates a remarkable capability in accurately identifying all invalid responses, evidenced by its complete absence of false-negative errors (the misclassification of invalid responses as valid).

R-V Guides generated by GPT-based methods also outperform those generated by Llama-based methods. Therapists point out that the major issue for Llama-based Guides is they sometimes get confused or distracted by the irrelevant follow-up answer and reorient from the original dimension to ask about contents mentioned in the

Table 2. The performance for **R-V Reasoner** (0: Valid, 1: Invalid), **R-V Guide**, and **R-V Validator** with different LLM models.

Prompted Models	R-V Reasoner			R-V Guide	R-V Validator
	Accuracy	Precision	Recall	Accuracy	Accuracy
GPT-4	97.8%	94.94%	100%	94.67%	96.79%
GPT-3.5 Turbo	97.3%	93.85%	100%	95.08%	95.95%
Llama-2-13b	69.3%	68.7%	84.5%	68.85%	83.47%
Llama-2-7b	74.54%	65.76%	70.50%	75.00 %	63.52%

follow-up. This phenomenon is more obvious in the Guides generated by Llama-2-13b. Therapists also observe that Llama-based Guides cannot capture all the points in the user response, often focusing only on one aspect.

Moreover, the empathic validations generated by the Llama-based **R-V Validators** are substandard in quality control. In particular, Llama-based **R-V Validators** perform poorly in following the objectives and format specified in the few-shot prompt. Some outputs included excessive small talk, some focused on follow-up responses that were irrelevant to the target dimension that was the primary focus of the discussion, and some failed to provide reflection or affirmation. For example, one output generated by Llama-2-13b **R-V Validator** engaged in questions aiming to problem solve based on the user’s responses, rather than offering empathetic validation as intended, which deviates significantly from the expected function of providing empathetic support. As indicated in Table 2, GPT-based **R-V Validators** deliver more regulated and superior quality empathic validations, largely conforming to the prompt’s requirements. However, a minor number of empathic validations by GPT-based **R-V Validators** encounter issues such as the use of inappropriate words or tones or overinterpreting the feelings and emotions in user responses.

5.4 Cognitive Behavioral Therapy Reasoner and Guide

As described in Section 2.2 and Section 4.2, the CBT process usually includes four steps and CaiTI completes the first step – identifying the situation and issues – for the user based on the historical user responses in the current conversation session. As such, there are three stages remaining: recognizing the negative thoughts (CBT_Stage1), challenging the negative thoughts (CBT_Stage2), and reframing the thoughts and the situations (CBT_Stage3). As the three stages target different objectives, there is a *Questioner*, *Reasoner*, and *Guide* in each stage.

Figure 9 shows the pipeline for the three-stage CBT process. In each stage, the **Questioner** poses questions aligned with that stage’s specific objectives. The **Reasoner** then evaluates the user’s response for its validity and utility. CaiTI progresses to the next stage if the response is deemed valid. If not, the **Guide** aids the user in crafting a valid response by identifying issues in the current response and suggesting possible improvements for a valid answer. If the user does not provide a valid response after two attempts, CaiTI will conclude the CBT process and recommend that the user seek professional assistance for a more effective and valid CBT experience.

Therapists also point out that an acceptable response involves identifications of cognitive distortion, such as polarized thinking, overgeneralization, emotional reasoning, catastrophizing, and jumping to conclusions. The **Reasoner** is tasked with recognizing responses containing cognitive distortions as valid, especially for **CBT_Stage1 Reasoner**. Meanwhile, if the response with cognitive distortions is invalid (e.g., not relevant to the situation), the **Guide** must take these distortions into account when assisting the user in formulating a valid response.

5.4.1 Microbenchmark – Cognitive Behavioral Therapy Reasoner and Guide. Psychotherapists provide us with 146 example situations within the 37 dimensions and the three-stage CBT responses for each situation. In addition, they label the validity of each response (0: Valid, 1: Invalid). In particular, there are 15, 17, and 33 responses that are invalid and need “Guide” in CBT_Stage1, CBT_Stage2, and CBT_Stage3, respectively. With guidance from therapists, we developed system content prompts to evaluate whether users can identify, challenge, and reframe

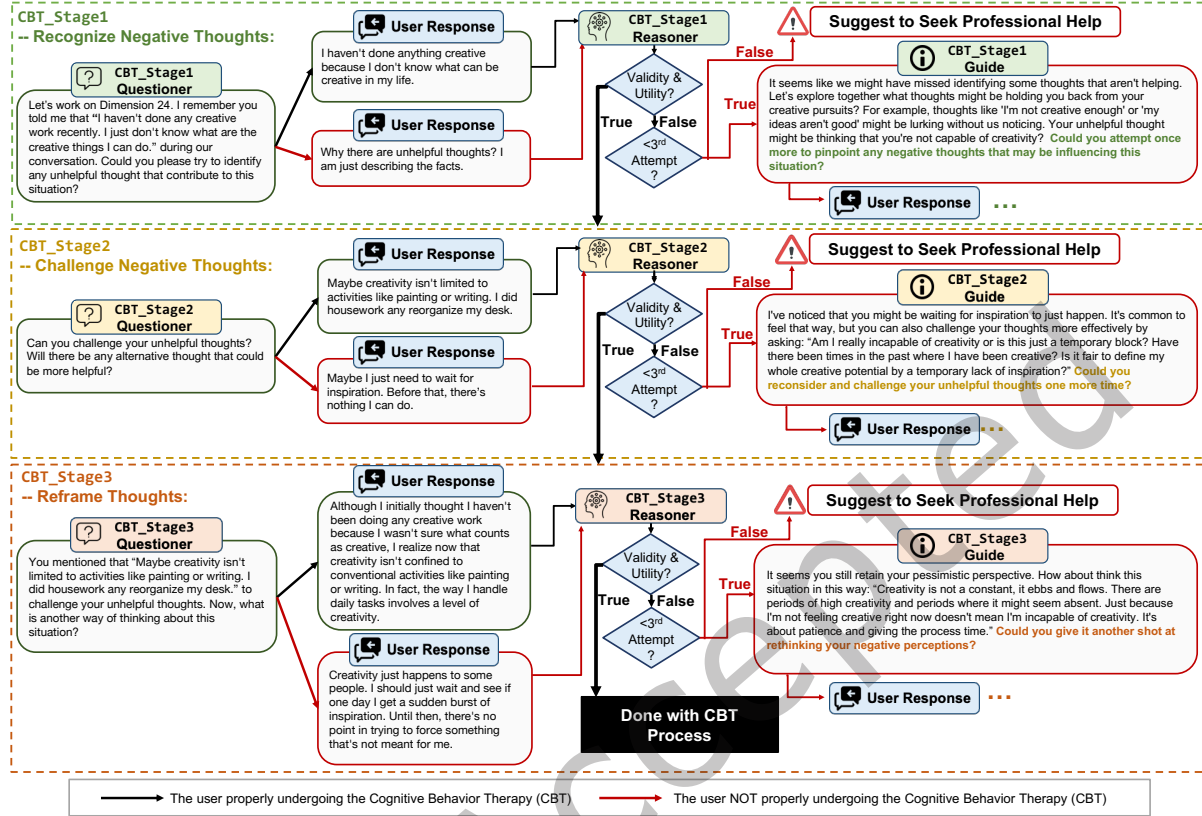


Fig. 9. The pipeline of the CBT process, which includes **Questioners**, **Reasoners** and **Guides**. The CBT process in CaiTI contains three stages: CBT_Stage1 – Recognize Negative Thoughts, CBT_Stage2 – Challenge Negative Thoughts, and CBT_Stage3 – Reframe Thoughts. The example scenarios where the user properly undergoes and fails to undergo each CBT stage are presented.

Table 3. CBT Reasoner performance for using different LLM models (0: Valid, 1: Invalid).

Prompted Models	CBT_Stage1 Reasoner			CBT_Stage2 Reasoner			CBT_Stage3 Reasoner		
	Accu-racy	Preci-sion	Recall	Accu-racy	Preci-sion	Recall	Accu-racy	Preci-sion	Recall
GPT-4	95.89%	72.68%	93.33%	95.21%	72%	100%	99.32%	100%	96.97%
GPT-3.5 Turbo	95.21%	83.33%	66.67%	94.52%	90.91%	58.82%	95.89%	100%	82.35%
Llama-2-13b	67.81%	23.33%	93.33%	91.1%	59.09%	76.47%	96.58%	96.67%	87.88%
Llama-2-7b	69.86%	10.81%	26.67%	88.36%	50%	29.41%	81.51%	58.33%	63.64%

negative thoughts effectively for CBT_Stage1 Reasoner, CBT_Stage2 Reasoner, and CBT_Stage3 Reasoner. And the performances of these CBT Reasoners using different LLMs are listed in Table 3. The user content for each stage encompasses all responses from the user and CaiTI within the current stage as well as from preceding stages. Overall, the performance for CBT_Stage3 is higher. GPT-3.5 Turbo has comparable performance to GPT-4, while both GPT-based models significantly outperform the Llama-2-based models in reasoning.

Table 4. **CBT Guide** accuracies for using different LLM models.

Prompted Models	CBT_Stage1 Guide	CBT_Stage2 Guide	CBT_Stage3 Guide
GPT-4	93.33%	94.12%	90.9%
GPT-3.5 Turbo	100%	100%	96.97%
Llama-2-13b	73.33%	94.12%	100%
Llama-2-7b	73.33%	100%	93.93%

As mentioned before, **CBT Guide** needs to be aware of cognitive distortions and point them out to guide effectively. According to psychotherapists' experiences, identifying unhelpful thoughts is the most challenging stage within the three stages of the CBT process, with the highest probability of containing cognitive distortions and less information obtained from the user in this stage. Table 4 shows the performance of different LLMs in providing appropriate **CBT Guide** when responses are invalid in the three CBT stages. In general, using the same prompt, GPT-3.5 Turbo performs the best, even outperforming GPT-4. Therapists comment that "*GPT-4 sometimes sounds like it is reading into the user's feelings*" instead of guiding the user objectively. Additionally, Llama-2-13b and Llama-2-7b show fairly good performance in **CBT_Stage2** and **CBT_Stage3**, with very few instances of "overanalyzing" or "reading into emotions". However, as previously mentioned, Llama models have comparably poor reasoning capabilities and struggle with processing more complex information, which might contribute to their inferior performance in **CBT_Stage1**.

5.5 Discussion and Observations on different LLMs

As illustrated in Sections 5.2 to 5.4, GPT-based models generally outperform Llama-based models in CaiTI. They better achieve the desired functionalities and adhere to the instructions specified in few-shot prompts, especially in complex tasks such as: (i) **Response Analyzer**, which classifies user responses into (**Dimension**, **Score**) with 5 general response classes, 37 dimensions, and 3 scores; and (ii) various **Reasoners**, **R-V Guide**, and **R-V Validator**, where user responses can cover an infinitely diverse spectrum and may deviate from the current conversation topic. Llama-based models struggle to follow the instructions in few-shot prompts when user expressions lack logical consistency or contain cognitive distortions.

Moreover, GPT-based models sometimes add their own interpretation of users' feelings instead of providing an objective, matter-of-fact output based on the user responses. In contrast, Llama-based models demonstrate more stable performance with few-shot prompts for **CBT_Stage2 Guide** and **CBT_Stage3 Guide**, where user responses are more standard and controlled. This stability is thanks to the filtering by **CBT Reasoners**, and because the tasks, which involve challenging and reframing negative thoughts, are more straightforward. These observations are expected, as GPT-based models have larger parameters and are pre-trained on a larger corpus of data. Indeed, there are methods that can further improve the quality of the outputs generated by these LLM models, which will be further discussed in Section 9.

6 IMPLEMENTATION AND STUDY DESIGN

In this section, we outline the subject recruitment procedures, describe the implementation, and detail the study design.

6.1 Subject Recruitment

20 subjects voluntarily participated (received informed consent from each subject) in our study (approved by the Institutional Review Board), including 10 men and 10 women between 18 and 40 years old from different races. All participants reported having normal hearing and cognition with no history of serious mental or physical

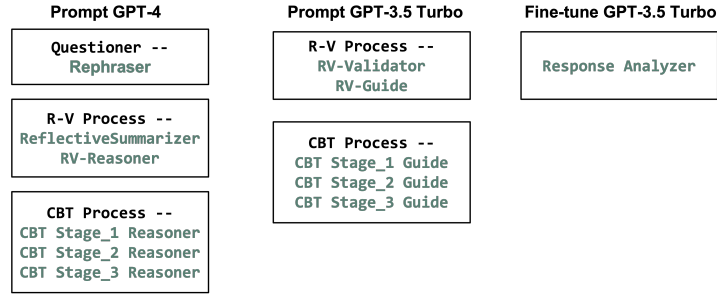


Fig. 10. The LLM implementation for different tasks during the study.

illness. All subjects were either students or employed. Each subject was assigned a random subject ID (e.g., S01) for data identification.

6.2 LLMs Implementation

Considering the response time and computational resources requirement, the best performing LLM according to the microbenchmarks is implemented for different modules, as illustrated in Figure 9. As discussed in Section 5, different LLMs excel at handling different tasks in CaiTI's conversation flow. For example, GPT-3.5 turbo is more suitable for providing guidance and empathic validation, while GPT-4 is better at reasoning the validity and utility of user responses.

6.3 Implementation

As mentioned in Section 1, to accommodate individuals with different needs and with possible memory or vision impairments, particularly the elderly, CaiTI is available in two physical form factors: a customized multi-platform app and a smart speaker (Amazon Alexa). CaiTI integrates custom Alexa Skills with an Amazon Echo device and a Flutter-based application, enabling flexible interaction through voice or text on multiple platforms such as Android, iOS, Windows, and macOS.

Figure 3 depicts the System's architecture, highlighting the communication pathway where only the conversation text is transmitted between the user interfaces and the server. This design decision is implemented to mitigate the risk of compromising sensitive information that may be inherent in voice data. Voice interactions are facilitated by APIs such as the Alexa Skills Kit [5] and Google's speech-to-text API [2]. The server's role is to handle all LLM-based tasks, including interpreting text inputs and generating appropriate follow-up questions or psychotherapeutic interventions that are then delivered to the user interface.

Figure 6 shows the smartphone interface for CaiTI's conversation session, where the various psychotherapeutic interventions applied during different stages of the conversation are annotated. Figure 11 displays the home page of the CaiTI on a smartphone (Figure 11a), shows user interactions with CaiTI via voice commands using both the smartphone and an Amazon Echo during in-lab sessions (Figures 11b and 11c), and depicts a user at home interacting with the CaiTI through text input on a computer (Figure 11d).

6.4 Study Design

We first conducted a 14-day study with in-lab and at-home sessions for each subject. The subjects participated in the in-lab session on the first and last days. Afterward, 4 subjects voluntarily participated in a 24-week at-home longitudinal study. A licensed psychotherapist implemented bi-weekly PHQ-9 and GAD-7 assessments with these 4 subjects to measure the effectiveness of CaiTI. Figure 12 shows the modules and tasks implemented in each setup. Subjects were told they were able to unselect some dimensions if they felt uncomfortable through the

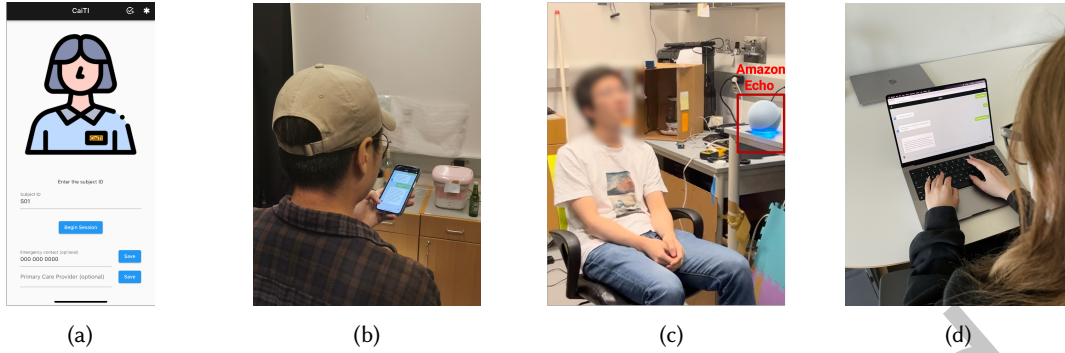


Fig. 11. (a): Home page of CaiTI's smartphone platform. (b) and (c): Subject chat with CaiTI via the smartphone/Amazon Echo at the end of an in-lab session. (d): A subject chats with CaiTI on her computer at home.

smartphone interface shown in Figure 7, but they were encouraged to select all. All conversation sessions (user and CaiTI responses, **Response Analyzer's** results, **Reasoners'** results) are saved for evaluation purposes.

6.4.1 In-Lab Session. The scope of the study and a tutorial about CaiTI system are given at the first in-lab session after informed consent was obtained. During this session, subjects were informed that the dialogue with CaiTI would incorporate elements of psychotherapy. However, to prevent their responses from being influenced and to maintain uniformity in the user experience, the underlying principles and methodologies of the psychotherapies (Motivational Interviewing (MI) and Cognitive Behavioral Therapy (CBT)) were not disclosed. Then, each subject is asked to choose their favorite method (smartphone platform/laptop platform/Amazon Echo) to converse with CaiTI. At the end of every in-lab session, subjects evaluated the system and provided feedback. This evaluative data will be examined in detail in Section 8.

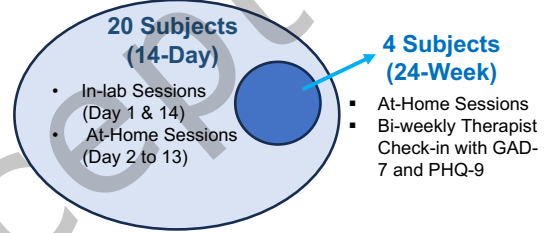


Fig. 12. The modules and tasks involved in the 14-day study and 24-week longitudinal study.

6.4.2 At-Home Session. Participants were requested to engage in dialogues with the CaiTI at their convenience, with a recommended frequency of once daily or, at a minimum, twice weekly. Among the cohort, four subjects agreed to extend their participation to a 24-week duration. These individuals had regular sessions with a licensed therapist throughout the study period. To evaluate the severity of depression and generalized anxiety disorder, the PHQ-9 and GAD-7 scales were administered bi-weekly, respectively. Psychotherapists closely monitored the participants to ascertain that involvement in the study did not exert any adverse effects on their daily lives.

7 SYSTEM EVALUATION

We combine all the logs from conversation sessions during the 14-day and 24-week study and evaluate CaiTI's system performance in this section. There are 454 conversation sessions with 18,309 and 2,013 segments of subjects' responses to CaiTI's original and follow-up questions, respectively. Some subjects answered the question with more than one sentence (segment). There are 107 times when the subjects failed to provide valid follow-up responses at the first attempt and triggered **R-V Guide**. CaiTI provides 2,013 empathic validation and support sessions with **R-V Validator**.

As described in Section 5.4, CaiTI will conclude the CBT process, if the user does not provide a valid response after two attempts in each stage of the three-stage CBT process. There are 454 CBT sessions at the end of each

Table 5. Number of times that **CBT Reasoner** and **CBT Guides** triggered in each attempt and the number of CBT sessions terminated in the three CBT stages.

	CBT_Stage1	CBT_Stage2	CBT_Stage3
CBT Reasoner in Attempt 1	454	451	448
CBT Guide in Attempt 1	23	37	21
CBT Reasoner in Attempt 2	23	37	21
CBT Guide in Attempt 2	10	7	5
CBT Reasoner in Attempt 3	10	7	5
Terminated in Attempt 3	3	6	3

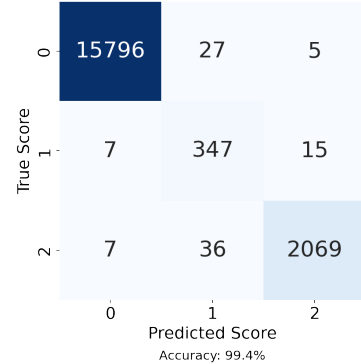
Table 6. The performance for **R-V Reasoner** (0: Valid, 1: Invalid), **R-V Guide**, and **R-V Validator** in handling the responses from subjects during the experiment.

	R-V Reasoner		R-V Guide	R-V Validator
Accuracy	Precision	Recall	Accuracy	Accuracy
97.97%	70%	98.99%	95.32%	96.57%

conversation session. Among these CBT sessions, as shown in Table 5 3, 6, and 3 of the CBT sessions terminated (fail to provide valid or relevant responses within 3 attempts) in **CBT_Stage1**, **CBT_Stage2**, and **CBT_Stage3**, respectively. The number of user attempts in each CBT stage is shown in this table. As such, there are 33 **CBT_Stage1 Guides**, 44 **CBT_Stage2 Guides**, and 26 **CBT_Stage3 Guides**, respectively. And the **CBT_Stage1 Reasoner**, **CBT_Stage2 Reasoner**, and **CBT_Stage3 Reasoner** are called 487, 495, 474 times, respectively.

4 licensed psychotherapists label the ground truth of output generated by these **Reasoners**, **Validator**, and **Guides** to evaluate their performance. Specifically, each therapist individually labels the outputs, with the ground truth determined by the majority vote among their evaluations.

In general, during the study with subjects, the performances of various LLM-based functional modules were either better than or comparable to their performances on the datasets provided by psychotherapists during the microbenchmark experiments. This phenomenon is expected since there is a lower probability for subjects to provide invalid or illogical responses compared to the proportion of “invalid” or “inappropriate” responses in the datasets provided by the therapists. *Note that CaiTI is designed for precautionary screening, assistance, and conversational psychotherapeutic intervention, and it is not intended to replace the process of diagnosis or clinical treatment.*



7.1 Response Analyzer

The user responses to the original questions were divided into segments. Each response segment was classified into **(Dimension, Score)** pairs, with ground truth labeled by the therapists. Dimension classification accuracy (5 classes of general responses and 37 dimensions) reached 97.6%. Figure 13 presents the confusion matrix for the **Score** of the 7,989 segments with a 99.4% accuracy.

Fig. 13. CaiTI’s performance to assign **Score** to the *Segments* of the user responses.

7.2 R-V Reasoner, R-V Guide, and R-V Validator

During the 2,013 reflection-validation (R-V) process, **R-V Reasoner** and **R-V Guide** were activated 2,120 and 107 times, respectively. As shown in Table 6, CaiTI’s **R-V Reasoner** almost perfectly identifies all “invalid” follow-up

Table 7. **CBT Reasoner** and **CBT Guide** performance in handling the responses from subjects during the three-stage CBT process (0: Valid, 1: Invalid).

CBT_Stage1 Reasoner			CBT_Stage2 Reasoner			CBT_Stage3 Reasoner		
Accuracy	Precision	Recall	Accuracy	Precision	Recall	Accuracy	Precision	Recall
99.38%	93.39%	96.88%	98.78%	88.64%	97.5%	99.57%	100%	92.85%
CBT_Stage1 Guide			CBT_Stage2 Guide			CBT_Stage3 Guide		
Accuracy			Accuracy			Accuracy		
90.9%			95.45%			92.23%		

responses from the user with only 1 exception. **R-V Reasoner** misclassified 42 valid follow-up responses as invalid, which is acceptable in the context of precautionary psychotherapeutic intervention, as it would guide the user to provide valid follow-up responses with better quality. In addition, **R-V Guide** achieved an accuracy of 96.57%, with only 5 guides being slightly not perfect. The causes for these 5 imperfect guides are overinterpreting the relationship between the follow-up response and the original response and missing some information from the user responses.

The therapists also checked all the 2,013 empathic validations and supports provided by the **R-V Validator** to the subjects through a majority vote, with 69 being slightly inappropriate. Specifically, when follow-up responses were too brief (under 3 words), CaiTI struggled to comprehend, leading to 24 improper empathic supports. 15 inappropriate instances resulted from inconsistencies between the subjects' original and follow-up responses. Another 30 inappropriate validations are due to the GPT-based **R-V Validator** adding its own interpretation of user responses into the empathic validation. Overall, CaiTI effectively delivered empathic validation and support in over 96.5% of instances. Although there are concerns about bias in large language models, the fine-tuned models in this work perform well in user studies and exhibit minimal bias.

7.3 CBT Reasoner and CBT Guide

Table 7 illustrates the performance of **CBT Reasoners** and **CBT Guides** when handling the responses from subjects from all attempts during the three-stage CBT process. It is shown that **CBT Reasoners** in each stage have high accuracies in identifying the validity and utility of the user responses to meet the psychotherapeutic goal of the CBT process. The **CBT Reasoners** also achieve high recall, which demonstrates that it is extremely rare for **CBT Reasoners** to miss any user responses that are invalid or unrelated.

CBT_Stage1 Guide, **CBT_Stage2 Guide**, and **CBT_Stage3 Guide** generated only 3, 3, and 2 less ideal contexts to guide the subjects. The most common issue for these 8 suboptimal **CBT Guides** is that they tried to read minds and made excessive assumptions about the relationships of the user responses earlier in the CBT process. None of these suboptimal **CBT Guides** completely misguide the user or lead them in the wrong direction.

8 USER STUDY

In this section, we present the quantitative and qualitative feedback of the 20 subjects who participated in our study (20 subjects participated in the 14-day study and 4 subjects extended to 24 weeks). We also organize and discuss the qualitative feedback and evaluations from the 4 psychotherapists working with us on this project.

8.1 User Adaptation Indicated by Device Usage

During the 24-week study, we meticulously tracked the number of conversation sessions each subject had per week with CaiTI. As illustrated in Figure 14, the data indicate that all four subjects engaged in dialogue with CaiTI with a frequency of 2 to 3 times daily. Notably, this consistent engagement over the course of the study

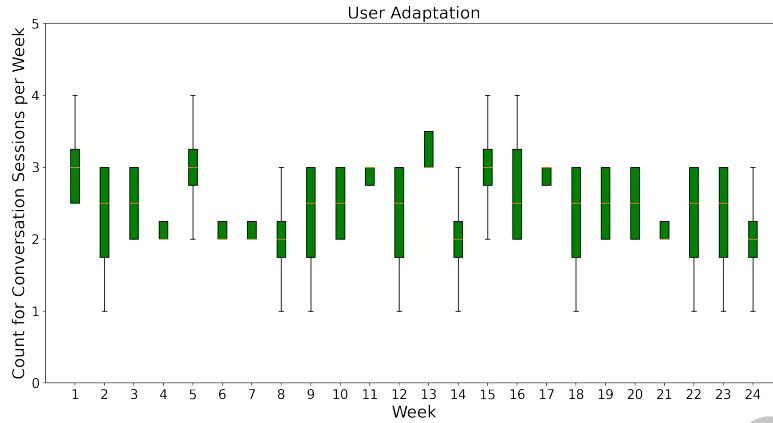


Fig. 14. The user adaptation CaiTI during the 24-week study.

suggests a sustained use of the system by the participants. The frequency of their interactions with CaiTI did not exhibit a significant decline over time, implying a stable user adaptation and a persistent incorporation of the system into their daily routines. This enduring engagement underlines the utility and user-friendliness of CaiTI, as well as its potential to maintain user interest and interaction over extended periods.

8.2 Quantitative Analysis

ChatBot Usability Scale (BUS-15) is a recently developed tool to assess end-users' satisfaction with chatbots [17]. Taking BUS-15 into account, we devised the 10 aspects listed in Table 8. Subjects rated the system from 1 (poor) to 5 (excellent) for these 10 aspects on the first and last day of the experiment. In general, subjects gave high ratings to CaiTI. Most subjects thought positively of the conversations with CaiTI and were willing to recommend and continue using CaiTI in the future. Only one subject showed a slightly negative attitude towards CaiTI and gave low scores. This subject thought that *"the technology is too intelligent and makes me worried that AI might be playing a too important role in my daily life"* (S19).

A Wilcoxon signed-rank test, a nonparametric statistical within-subject test commonly used in behavioral science [81], was performed to compare the subjects' day-to-day functioning on the first and last days of the 14-day study based on their responses **Score** on all 37 dimensions (**Score** $\in \{0, 1, 2\}$, described in Section 4). The test results show a statistically significant decrease in the dimensions with **Score** equal to 1 and 2 on the last day, as compared to the first day of the experiment ($z = -2.68$, $p < .01$). This suggests an overall reduction in the dimensions that subjects reported as having functional concerns. The results indicate that the subjects' day-to-day functioning may have improved during the experiment using CaiTI.

In a 24-week longitudinal study, 4 subjects completed additional GAD-7 and PHQ-9 assessments every two weeks. These tools evaluate the severity of anxiety and depression symptoms. With the small sample size ($n=4$), therapists reviewed the questionnaire results individually. During the study, two subjects improved from moderate-to-severe anxiety and depression to mild levels, initially facing difficulties in functioning but eventually reporting none. One subject progressed from mild to minimal anxiety and depression and reported no functional difficulties. The remaining participant exhibited fluctuating symptoms, ranging from mild to minimal, initially facing some difficulties in functioning but reporting none by the study's end. The findings of this study indicate that CaiTI was effective in reducing the severity of anxiety and depression, as well as enhancing daily functioning.

Table 8. The average ratings on the first day (Day 1) and last day (Day 14) of the study.

Question Asked	Average Rating	
	Day 1	Day 14
How do you like CaiTI in general?	4.3	4.45
Are you willing to continue using CaiTI in the future?	4.3	4.35
Will you recommend CaiTI to others?	4.2	4.55
Do you think CaiTI's functions are easily detectable?	4.4	4.5
Do you think CaiTI can understand you and chat with you naturally?	4.35	4.5
Do you like the interface design of CaiTI?	4.6	4.2
Do you think the wording used in CaiTI is appropriate?	4.6	4.75
Do you think the psychotherapies are helpful?	4.65	4.6
How do you feel about the response time from CaiTI?	4.8	4.85
Do you feel that the conversation flow of CaiTI is gradually tailoring to you?	N.A.	4.75

8.3 Qualitative Evaluation from the Subjects and Therapists

8.3.1 Experiences and Feedback from Subjects. All 20 subjects who participated in the study found CaiTI to be **valid and effective**. Subjects see using CaiTI brings awareness to their mental health daily, which makes them feel they are “relying on the system to do the self-reflection work” (S10) every day. One of the subjects said: “I feel in good shape doing these check-ins every day, in between my weekly sessions with individual therapist” (S11). Subjects were also surprised about how well “CaiTI can understand me and makes me feel validated” (S17). One subject stated that:

“I think CaiTI does a pretty good job validating my feelings and encouraging me to be more active. I had a very positive experience with the comforting part. I also like the sensing part, as CaiTI knows what I am doing at home and directly provides help if needed. I don’t need to describe my daily routine and recall when I am with CaiTI. I need to do these annoying things during my therapist visits. I think CaiTI is a good add-on in mental healthcare” (S15).

Several subjects complimented on CaiTI’s **interactiveness** and stated “CaiTI is really plug-and-play and easy to use” (S01). Additionally, a large portion of subjects stated that they liked the way that CaiTI “talked” to them as “the conversation is very genuine” (S12). For example, one subject said:

“I feel like that I have a companion. When CaiTI talks to me through my Alexa, it listens and converses naturally and reminds me of parts of my therapy sessions. The system reformulates the questions it asks me every time. Also, I realize that if I use CaiTI more frequently, CaiTI is more attuned to me because it changes the way it asks the question according to my answers” (S02).

Subjects found the **psychotherapies helpful**, encouraging, and valuable. A subject said “I like the counseling part of the system. The tone is supportive and encouraging. The system really understands what I said to it and provides reasonable and applicable guides” (S08). In addition, a few subjects find the guidance (**Guide** feature) provided by CaiTI during the psychotherapies helpful.

“To be honest, I was not that familiar with what they called the CBT procedures at the end of each conversation. Initially, I did not know what are the “unhelpful thoughts” in my situations. But, you know, after hanging using CaiTI a few times, I started to get the hang of these helpful thinking strategies. I am getting more optimistic and better at boosting my own confidence, when I face challenges.” (S01).

A few subjects feel that “*the consolation is not pointed enough*” (S02), as it provides very general comments that are suitable for “*everyone who has the same problem*” (S02). They would like to have more personalized experiences with more targeted suggestions.

8.3.2 Comments from Therapists. 4 psychotherapists approved of CaiTI, recognizing its potential for “*combining physical and mental wellness screening and providing psychotherapeutic precautionary interventions in daily life for everyone*” (T02). They provided positive feedback on the conversational daily functioning screening, the integration of psychotherapeutic techniques within the conversation flow, and the overall style of the dialogue. They **approved the content and style of the system’s language**: “*CaiTI engaged me in conversations with effective content control. However, ChatGPT sometimes produced responses that are not up to clinical standards*” (T01).

The therapists also saw the interventions as **potentially useful in addition to clinical treatment**. They spoke positively about the empathic validation and support in the MI process and valued the significant influence of CaiTI’s **Reasoner** and **Guide** features in guiding and steering users toward more adaptive thinking in the CBT process. They felt that the combination of reasoning and guidance not only prevents users from getting stuck in incorrect ways of thinking but also positively directs and facilitates gradual learning. This, in turn, can support overall mental health well-being, especially in the long term. They mentioned these designs could “*enforce the effectiveness of CaiTI’s psychotherapy*” (T03) and “*provide just the right amount of guidance to support users in situations that require additional attention*” (T01). One therapist stated:

“From my observations on the interactions between CaiTI and our subjects, I think the validation implementations of CBT and MI are appropriate. Although very few exchanges are not perfect if the responses from the subjects are ambiguous, CaiTI impresses me with emotionally supportive text formulation. I also answered questions in a way a few times to indicate daily functioning concerns to CaiTI during different sessions, and CaiTI came up with different ways to guide and help me to improve the situation.” (T02).

Most therapists found CaiTI to **aid traditional psychotherapeutic processes**, stating that **more frequent data collection offers more insights** between therapy sessions. “*I would love to see some of my clients use this*” (T04), one therapist commented. Another therapist mentioned:

“Doing everyday check-ins helps people in general to bring awareness to mental health in their everyday activities. Sometimes, I assign my clients homework to log their everyday activities, just for them to keep doing the work outside of therapy sessions. It would be great if more people can use this system daily and become more intentional in their daily routine” (T02).

They also validated CaiTI’s use of reinforcement learning for conversation generation and “*are surprised by how attentive CaiTI is and how good the flow of the conversation we have*” (T04). Additionally, they offered suggestions for future enhancements, including adjustments to CaiTI’s audio tone.

9 DISCUSSION AND FUTURE WORK

As shown in Sections 7 and 8, CaiTI can effectively perform *Converse with the User in a Natural Way*, and provide *Psychotherapeutic Conversational Intervention*. The qualitative evaluations further attest to the system’s usability, the relevance and effectiveness of its psychotherapeutic content, the helpfulness of the guides, and the efficacy of the empathic validation it offers, making it a promising tool for personalized mental health care. Throughout the rest of this section, we summarize some of the current limitations and propose plans for future improvements and visions.

First of all, we will continue our collaboration with the psychotherapists. We plan to include real patients with different kinds and severities of mental disorders in longitudinal studies. We will evaluate how well CaiTI can assist the treatment provided by the therapists and improve the mental well-being of the patients.

Moreover, during the psychotherapy process, although CaiTI breaks the tasks down and leverages LLM-based **Reasoners**, **Guides**, and **Validator** to specifically handle each subtask, there is room to improve the accuracies of **Reasoners** and quality of **Guides**, and **Validator**. In particular, we plan to add step-by-step “system reasoners” (Chain-of-Thought) to evaluate if the **Guides** and **Validators** are suboptimal because of “reading users’ minds” and making excessive assumptions (i.e., hallucination) [71]. We also intend to implement ethical guidelines and governance measures for our LLM-based modules to ensure that the outputs are controlled and free from harmful content.

Additionally, we plan to investigate if further breaking down the tasks would improve the system’s performance. For example, we will investigate whether using two LLM modules for classifying **Dimension** and **Score** in **Response Analyzer** instead of one will improve the accuracy and how this change affects the performance of different LLM models. We are aware that these modifications will increase the computational overhead for the system, yielding longer system response times and affecting the user experience. As such, we also plan to investigate the trade-off between system complexity and user experience across multiple aspects.

We see the potential of incorporating common wearable devices, such as smartwatches and smartphones, as potential platforms into CaiTI. In fact, around half of the subjects in this study actively use smart devices for health and fitness monitoring. We plan to leverage the health and fitness data analyzed by smartphones or smartwatches (e.g., Apple HealthKit data) as a source to perform activity detection with existing commercial smart home devices.

In addition, the smart home market is rapidly expanding with new sensors and devices. CaiTI has the potential to be one of the many applications commonly integrated into smart home ecosystems. As such, we plan to investigate how to take advantage of smart home sensors, devices, and robots to provide a more comprehensive and invasive screening of users’ daily functioning through smart sensors and various kinds of interactions and interventions through home robots or other devices. With omnipresent modules being included in CaiTI, to make users feel less “invasive”, we plan to make CaiTI plug-and-play. Users can turn on devices that make them feel comfortable at that moment, CaiTI can automatically discover available resources and generate execution pipelines to screen the wellness of the user and provide interventions if necessary.

Furthermore, equipping with the RL recommender as well as fine-tuned and few-shot prompted GPT-based models for conversation, conversational psychotherapeutic intervention generations, CaiTI can generate speech for conversation and lead the conversation flow in a more human-like way compared to other platforms. However, since CaiTI uses the text-to-speech API, the tone and inflection of the voice generated to “talk” to the user are still not entirely identical to what a real person would perform. A therapist commented: *“It’s not exactly how a real person would speak in terms of the tone and inflection but I think that definitely will improve in time as you know text-to-speech and other natural language things become better and better. However, the content and style of the sentences and words are very much in line with what a typical person would say”* (T04). We plan to investigate and incorporate deep learning methods to add “emotion” to the audio output accordingly [8].

We have designed CaiTI to minimize bias by implementing several modules of LLMs tailored to specific tasks instead of relying on a single model. We acknowledge that despite our best efforts, all AI applications, including this one, are subject to potential bias, and the ethical concerns of AI for psychotherapies still remain. Therefore, the intended application of CaiTI is primarily for precautionary day-to-day functioning screenings and psychotherapeutic interventions, aiming for better self-care and assisting the professional psychotherapy process.

10 CONCLUSION

In this work, collaborating with licensed psychotherapists, we propose CaiTI, a conversational “AI therapist” that leverages the LLMs and RL to screen and analyze the day-to-day functioning of the user across 37 dimensions

and provides appropriate and effective psychotherapies, including MI and CBT, depending on the physical and mental status of the user through *natural* and *personalized* conversations. Accessible on common smart devices like smartphones, computers, and smart speakers, CaiTI provides a versatile solution for users, both indoors and outdoors.

To enhance care quality and psychotherapy effectiveness while minimizing AI biases, CaiTI employs specialized LLM-based **Reasoners**, **Guides**, and **Validator** to provide psychotherapy during the conversation. By using conversation datasets annotated by licensed psychotherapists, we assess various GPT and Llama-2 LLMs' performances through few-shot prompts or fine-tuning across different LLM-based tasks in CaiTI. Based on the assessment results and observations, we collaborate with the therapists to develop a proof-of-concept prototype of CaiTI and conduct 14-day and 24-week studies in real-world trials. The evidence gathered through our studies showcases CaiTI's effectiveness in screening daily functioning and providing psychotherapies. Additionally, the empirical evidence underscores CaiTI's capability in enhancing daily functioning and mental well-being. We see the potential for LLM-based conversational "AI therapists" to aid the traditional psychotherapy process and provide precautionary mental health and psychotherapeutic self-care to a larger population.

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A DIMENSIONS FOR DAY-TO-DAY FUNCTIONING SCREENING

This section presents the 37 dimensions proposed by [63] to screen day-to-day functioning. These dimensions are based on the Diagnostic and Statistical Manual of Mental Disorders (DSM)-IV and the Daily Living Activities–20 (DLA-20), two gold standards for mental health care providers. As mentioned in Section 5, for each dimension, therapists provide a set of sample questions. With the GPT-4-based Rephraser, CaiTI rephrases and formulates the questions when it converses with the user. Note that as some of the dimensions might be sensitive or uncomfortable for users, CaiTI offers the option for users to select the dimensions to work on manually through the smartphone interface. The 37 dimensions and the example questions asked by CaiTI are listed below:

- (1) *Maintaining stable weight*: “Have your weight changed significantly recently?”;
- (2) *Managing mood*: “How’s your mood recently?”;
- (3) *Taking medication as prescribed*: “Have you been taking medication according to your prescriptions?”;
- (4) *Participating primary and mental health care*: “Have you been seeing your doctor, therapist or case manager consistently?”;
- (5) *Organizing personal possessions & doing housework*: “Have you been doing house chores?”;
- (6) *Talking to other people*: “Have you been talking to other people?”;
- (7) *Expressing feelings to other people*: “Have you expressed feelings towards others?”;
- (8) *Managing personal safety*: “Have you been taking safety into consideration when making decisions?”;
- (9) *Managing risk*: “Have you taken any risk recently?”;

- (10) *Following regular schedule for bedtime & sleeping enough*: “How has your sleep been? Do you have a regular schedule for bedtime?”;
- (11) *Maintaining regular schedule for eating*: “How’s your eating? Are you eating regularly?”;
- (12) *Managing work/school*: “Are you showing up for work or school?”;
- (13) *Having work-life balance*: “How’s your work-life balance? Have you taken days off recently?”;
- (14) *Showing up for appointments and obligations*: “Are you showing up for appointments and other obligations?”;
- (15) *Managing finance and items of value*: “How’s your finances? Any concern with your spending habits?”;
- (16) *Getting adequate nutrition*: “How’s your nutrition? Are you eating healthy?”;
- (17) *Problem solving and decision making capability*: “Are you able to make decisions yourself?”;
- (18) *Family support*: “Do you feel supported by your family?”;
- (19) *Family relationship*: “How’s your relationship with your family members?”;
- (20) *Alcohol abuse*: “Do you often drink alone?”;
- (21) *Tobacco abuse*: “Do you smoke cigarettes or vape? And how frequent?”;
- (22) *Other substances abuse*: “Do you use any substance and what’s the frequency of using?”;
- (23) *Enjoying personal choices for leisure activities*: “What do you like to do when you have free time?”;
- (24) *Creativity*: “Have you done any creative work recently?”;
- (25) *Participation in community*: “What do you do in your neighborhood or community?”;
- (26) *Support from social network*: “Other than family members, who do you consider as your close support?”;
- (27) *Relationship with friends and colleagues*: “Do you hang out with friends or coworkers?”;
- (28) *Managing boundaries in close relationship*: “Do you feel comfortable with your partner or partners if you have?”;
- (29) *Managing sexual safety*: “If you are sexually active, do you try to avoid risky sexual behaviors?”;
- (30) *Productivity at work or school*: “Are you productive at work or school?”;
- (31) *Motivation at work or school*: “How’s your motivation for work or school?”;
- (32) *Coping skills to de-stress*: “What kind of coping do you use to calm yourself?”;
- (33) *Exhibiting control over self-harming behavior*: “Do you have self-harming behaviours?”;
- (34) *Law-abiding*: “Have you been arrested recently?”;
- (35) *Managing legal issue*: “Do you have any legal issue recently?”;
- (36) *Maintaining personal hygiene*: “How’s your personal hygiene? Are you taking care of your personal hygiene?”;
- (37) *Doing exercises and sports*: “Have you recently exercised?”;

As mentioned in Section 4.1, users can freely chat with CaiTI using open-ended responses, including both deterministic YES/NO answers and detailed responses. For example, when CaiTI asks: “Do you often drink alone?”, the user can respond deterministically: “Yes, I drink alone.” or: “I drink alone almost every other night recently.”. CaiTI classifies both responses to **Dimension Alcohol usage** with **Score** of 2. With **Score** = 2, CaiTI follows up and provides *Psychotherapeutic Conversational Interventions* starting by asking for more information. In addition, if the user says: “I don’t drink alone. But I like to drink with my friends when we hang out together.”, CaiTI segments the responses into two parts and classifies the first part to **Dimension Alcohol usage** with **Score** of 0, and the second part to **Dimension Relationship with friends and colleagues** with **Score** of 0.